Stage 2 Test Report: Heat-Welding & Adhesive Bonding Limit & Burst Report

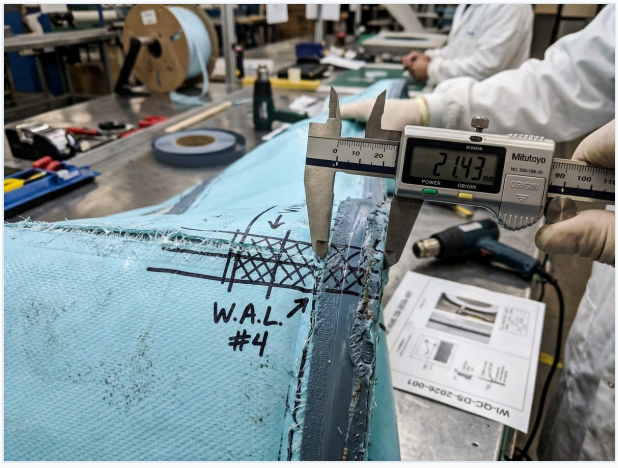
Extreme Thermal Creep & Step-Up Overpressure Burst Verification of Composite Seams on Felizbay Pool (400 cm × 300 cm × 120 cm, 20 cm Wall)

1. Manufacturing Traceability Archive & Test Scenario

Report No.	`RPT-FB-QC-20261113-S2`	**Governing SOP**	`WI-QC-DS-2026-001 (V1.0)` Stage 2
Brand / Model	**Felizbay** Large Rectangular Drop-Stitch Pool (`FB-DSP-400`)	**Dimensions / Wall**	**400 cm × 300 cm × 120 cm** / Wall Thickness **20 cm**
Composite Bonding	**RF / Hot-Air Heat Welding + 2-Part PU Cold-Bonding Adhesive**	**Rated Pressure**	**P_w = 8.0 PSI** (0.55 bar)
Sample Batch / Unit	Batch `LOT-202611-A08` / **Unit #2** (Full-Unit Burst Sample)	**Tape & Adhesive Lot**	80 mm Dual-Layer Tape `WB-261015-02` / PU Glue `PU-GL8820-B` (`GL-20261022`)
Machine & Operator	RF Welders `RF-04#`, `RF-07#` (ID `0182`) / Cold-Bond `CB-03#` (ID `0215`)	**Safety / Result**	Steel Blast Cage + ≥ 5 m Remote Control / **Stage 2 Approved (PASS)**

Test Scenario & Bonding Process Focus: Stage 2 subjects full-scale **Felizbay Unit #2** (400 × 300 × 120 cm, 20 cm wall) to extreme thermal-pressure creep and ultimate burst limits. Straight main seams and edge bands are joined by **RF / hot-air heat welding**, while 3D radiused corners, double-layer gusset patches, and valve flanges are reinforced via **MEK surface activation + two-component polyurethane cold-bonding adhesive**. Tests were executed in a 55°C environmental chamber and a steel blast-containment cage.

2. Stage 2 Test Setup & Composite Seam Inspection Photographs



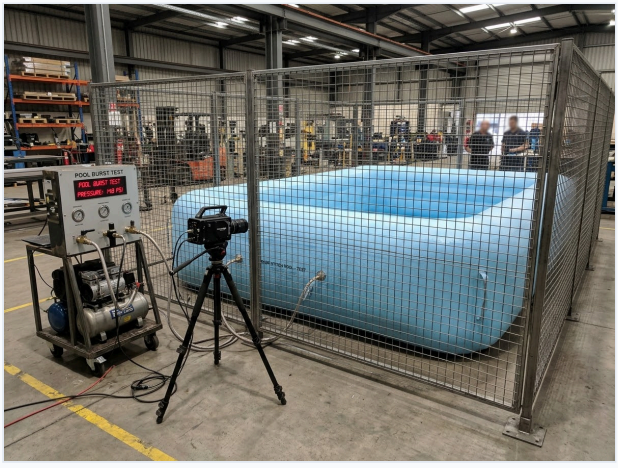


Fig. 2-1: Witness Alignment Line (W.A.L. #4) Seam Creep Check (Test 2.1)
Cross-seam alignment grids scribed every 200 mm across the heat-welded and PU adhesive edge band are verified via digital caliper after 48 h at 55°C / 12.0 PSI.

Fig. 2-2: Step-Up Burst Test & 480 fps Camera in Blast Cage (Test 2.2)
Unit #2 pressurized remotely from beyond the 5 m yellow safety line via the digital burst console while 480 fps high-speed cameras monitor corner adhesive patches and welded seams.

3. Detailed Test Parameters & Measured Data Records

3.1 Test 2.1: 55°C High-Temperature & 1.5×P_w Creep 48-Hour Endurance Test

Test Parameters: Inflated at 23°C to **P_w = 8.0 PSI** for 30 min; **10 crosshair thickness points** (T_0) marked across the 20 cm wall (4 mid-wall, 4 corner composite zones) and 10 cm floor (2 center), plus Witness Alignment Lines (W.A.L.) every 200 mm. Conditioned at **55°C ± 2°C** for 1 h, then pressurized to **1.5 × P_w = 12.0 PSI** for **48 continuous hours** (Spec: thickness expansion δ ≤ 10%, W.A.L. slip ≤ 1.0 mm, pressure drop ≤ 8%).

Monitored Zone & Seam Type	Initial Mean T_0 (mm)	24h Mean T_{24} (mm)	48h Mean T_{48} (mm)	Max Point Expansion δ (Spec ≤ 10%)	Max W.A.L. Slip (Spec ≤ 1.0 mm)	Result
Wall Mid-Span (4 pts, 20 cm)	200.25	206.80	209.10	**4.7%** (209.6 mm)	0.35 mm (Heat Weld)	**PASS**
Wall Corner Zone (4 pts, Weld+Glue)	200.40	207.50	210.25	**5.2%** (210.8 mm)	0.42 mm (PU Adhesive)	**PASS**
Floor Center (2 pts, 10 cm)	100.15	103.10	104.30	**4.3%** (104.5 mm)	0.25 mm (Heat Weld)	**PASS**

48-Hour Monitoring Notes: Chamber gauge pressure shifted from 12.0 PSI to 11.6 PSI (**3.3% drop ≤ 8%**); IR thermography showed ≤ 0.6°C delta across welded and adhesive seams; post-cooling soap-solution inspection confirmed 0 bubbles and zero localized drop-stitch delamination.

3.2 Test 2.2: Step-Up Overpressure Burst Test inside Blast-Containment Cage

Test Parameters: Cooled **Unit #2** locked inside the steel blast cage; personnel stationed behind the 5 m safety barrier. Starting at **12.0 PSI (1.5×P_w)**, pressure was stepped up by **+2.0 PSI** within 30 s per stage and held for **3 minutes** (5 min at Stage 1) until structural rupture under **480 fps** high-speed video and Peak Hold recording (Spec: P_{burst} ≥ 2.5 × P_w = 20.0 PSI).

Step	Target Pressure (PSI)	Multiple of P_w	Hold Duration	High-Speed Video & Composite Seam Observation	Status
Step 1–3	12.0 → 16.0	1.50 – 2.00 × P_w	5 min + 3 min×2	20 cm wall rigid without distortion; welded edge bands and PU corner patches silent	Normal

Step	Target Pressure (PSI)	Multiple of P_w	Hold Duration	High-Speed Video & Composite Seam Observation	Status
Step 4–6	18.0 → 22.0	2.25 – 2.75 × P_w	3 min / step	**20.0 PSI (2.5×P_w)** passed 3 min hold; faint fiber tension ticks heard at 22.0 PSI	Spec Met
Step 7	24.0	3.00 × P_w	3 min	Uniform elastic wall crowning; zero weld peeling or adhesive edge lifting during 3 min hold	Normal
Step 8	**25.4 (Peak)**	**3.18 × P_w**	Ruptured at 25.4 PSI	**First failure initiated in mid-panel drop-stitch yarns**; all RF/hot-air welds and PU cold-bonded corner reinforcements remained intact with zero cold-weld or over-weld cracking	**PASS**

4. Stage 2 Conclusion & Bonding Process Evaluation

Stage 2 Summary Conclusion: The hybrid **RF / hot-air heat welding + two-component polyurethane cold-bonding adhesive** process used on **Felizbay** (FB-DSP-400, 20 cm wall) withstood 48 hours at 55°C / 12.0 PSI (max thickness creep 5.2% ≤ 10%, max seam slip 0.42 mm ≤ 1.0 mm) and achieved an ultimate burst pressure of **25.4 PSI (3.18 × P_w)** with parent-fabric rupture rather than joint failure, confirming full process capability.

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